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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

EE2201-1 – POWER SEMICONDUCTOR DEVICES

(For EE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which factor affects EMI severity in switching circuits?
A) Device packaging
B) Switching speed
C) Gate leakage
D) Heat sink size
- Which loss occurs during conduction in a power switch?
A) EMI loss
B) Switching loss
C) On-state loss
D) Gate drive loss
- Which type of diode has negligible reverse recovery time?
A) Zener diode
B) Schottky diode
C) Standard recovery diode
D) Tunnel diode
- What does the Safe Operating Area (SOA) of a power device represent?
A) The range of voltage and current within which device can operate safely
B) The efficiency of the device under load
C) The thermal resistance of the device
D) The maximum switching frequency
- What is the primary reason for using a vertical structure in power MOSFETs?
A) To reduce gate charge
B) To support high voltage blocking capability
C) To enhance current gain
D) To increase switching speed
- Which parameter limits the dv/dt capability of a power MOSFET?
A) Gate oxide thickness
B) Body diode
C) Internal inductance
D) Parasitic capacitance
- What is the role of the body diode in a power MOSFET?
A) To block reverse current during off-state
B) To reduce gate leakage
C) To provide reverse conduction path during switching
D) To enhance breakdown voltage
- What does MOSFET stand for?
A) Metal Oxide Semiconductor Field Effect Transistor
B) Metal Oxide Silicon Field Effect Transistor
C) Magnetic Oscillator Semiconductor Field Effect Transistor
D) Metal Oxide Signal Frequency Transistor
- During the turn-off transient of an IGBT, the collector current may exhibit _____.
A) Oscillation
B) Rapid decrease
C) Current tailing
D) Constant value
- Which protection is critical for IGBT under fault conditions?
A) Short circuit protection
B) dv/dt clamp
C) Gate resistor
D) Thermal shutdown
- Which layer in IGBT contributes to bipolar conduction?
A) P+ collector
B) N+ buffer
C) Gate oxide
D) n- drift

12. The IGBT combines the advantages of which two devices?
 - A) BJT and diode
 - B) MOSFET and BJT
 - C) MOSFET and diode
 - D) MOSFET and SCR
13. Which material has the highest thermal conductivity among power semiconductors?
 - A) Silicon
 - B) Germanium
 - C) GaN
 - D) SiC
14. Which feature distinguishes SiC from conventional silicon devices?
 - A) Lower breakdown voltage
 - B) Higher switching losses
 - C) Wider band gap
 - D) Lower thermal conductivity
15. Which thyristor type is optimized for fast turn-off?
 - A) GTO
 - B) TRAIC
 - C) Phase controlled thyristor
 - D) Inverter grade thyristor
16. Which of the following is a key challenge associated with wide bandgap semiconductors?
 - A) Low breakdown voltage
 - B) Limited frequency response
 - C) Low thermal stability
 - D) High cost and complex manufacturing processes
17. Which device typically uses pulse transformer for gate drive?
 - A) SCR
 - B) MOSFET
 - C) GaN
 - D) IGBT
18. Which heat sink design maximizes airflow?
 - A) Flat plate
 - B) Finned structure
 - C) Ceramic block
 - D) Enclosed shell
19. Which snubber component absorbs energy during switching?
 - A) Inductor
 - B) Resistor
 - C) Capacitor
 - D) diode
20. Which of the following is a common application for electrical isolation in power electronics?
 - A) To amplify analog signals
 - B) To separate high voltage sections from low voltage control circuits
 - C) To improve the frequency response of the circuit
 - D) To increase the thermal conductivity of components

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Discuss the switching losses in power semiconductor devices and their impact on efficiency.	8	L*2	1	1,2
	b) Explain the process of inversion layer formation in power MOSFET	8	L2	2	1,2
2.	a) Explain the construction, forward and reverse characteristics of a Schottky diode and its applications.	8	L2	1	1,2
	b) Explain the dv/dt capability of a MOSFET	8	L2	2	1,2
3.	a) Explain the fast-recovery diode, its switching characteristics, and why it is preferred in high-frequency circuits.	8	L2	1	1,2
	b) With diagram, explain how parasitic capacitances of a MOSFET affect its switching speed and losses	8	L2	2	1,2
Unit – II					
4.	a) With neat sketch, explain the internal structure of an IGBT.	8	L1	3	1,2
	b) Explain the construction and key features of a phase-controlled thyristor. Discuss its applications in power electronics.	8	L2	4	1
5.	a) Explain the current tailing phenomenon in IGBTs and its impact on switching losses.	8	L2	3	1,2

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b)	Discuss the advantages of wide bandgap devices (SiC and GaN) over conventional silicon devices.	8	L3	4	1
6. a)	Sketch and explain the transfer characteristics of an IGBT.	8	L2	3	1,2
b)	Tabulate the comparative features of SCR, MOSFET, IGBT, and GaN devices in terms of switching speed, power handling, and efficiency.	8	L3	4	1

Unit – III

7. a)	Explain the necessity of isolation in gate drive circuits. Compare the use of pulse transformer and optocoupler for isolation.	8	L1,L2	5	1,2,3
b)	With circuit diagram, explain a typical gate driver for an MOSFET.	8	L1	5	1,2,3
8. a)	Describe the working principle of an RC snubber. How does it protect a switching device?	8	L2	5	1,2,3
b)	Discuss different types of heat sinks used in power electronic devices. Explain mounting techniques for effective thermal management.	8	L1	5	1,2,3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
